

Nahyun Lee

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Boston, MA, United States

RESEARCH INTERESTS

Augmenting how we perceive the world.

My research explores the intersection of Cognitive Augmentation, VR/AR, Haptic and Human-Computer Interaction. I aim to help people interpret an information-overloaded world.

EDUCATION

- **University of Massachusetts Boston** Sep 2026 – Jun 2027
Incoming Exchange Student Boston, MA, USA
- **Sungkyunkwan University** Feb 2023 – Expected Feb 2028
B.S. in Mechanical Engineering, Software Engineering, and Self-Designed Major in HCI Suwon, South Korea
◦ GPA: 4.34/4.50
- **Hanyang University** Mar 2022 – Dec 2022
Bachelor of Business Administration Coursework (First-Year Completed) Seoul, South Korea

RESEARCH EXPERIENCE

- **Wearable and Interactive Technology Lab, KAIST** Jan 2026 – Present
Research Intern, Advised by Prof. Ian Oakley Daejeon, South Korea
 - Investigating beamforming-based audio sensing to enhance perception of everyday information
 - Developing audio-driven granular synthesis for high-DOF vibrotactile feedback in haptic interaction
- **Smart Biomedical Lab, Sungkyunkwan University** Jun 2025 – Dec 2025
Research Intern, Advised by Prof. Jowoon Chong Suwon, South Korea
 - Developed a load cell- and IMU-based muscle function assessment device with BLE
 - Built a mobile app for real-time sensing and accessible interaction for older adults
 - Collaborated with Korea University Medicine and KAIST on muscle function assessment for older adults
- **Interaction Independence Study, Stanford Center at the Incheon Global Campus** Mar 2024 – Dec 2025
Research Assistant, Advised by Hoseok Jung Incheon, South Korea
 - Prototyped thermal haptic interfaces based on radiative and convective heat transfer
 - Developed a physics-based fire visualization pipeline using Fire Dynamics Simulator (FDS) for AR/VR
 - Collaborated with TU Wien and Imperial College London on thermal haptics for firefighter training

PUBLICATIONS

J=JOURNAL, C=CONFERENCE, W=WORKSHOP, D=DEMO, P=POSTER, U=UNDER REVIEW

- [U.2] Sara Scheffer*, Elias Panner, **Nahyun Lee**, Hoseok Jung, Wanting Mao, Sebastian Lux, Frank Long, Arnab Majumdar, and Kurt Matyas.
Embodied Multimodal Mixed Reality for Firefighter Training: Evaluating Visual-Thermal Integration for Situational Awareness and Decision Performance
- [U.1] Hoseok Jung*, Jiyeon Lee, **Nahyun Lee**, and Hyunmin Kang.
Real-Time Thermal Haptics for Fire Simulation Modeling Radiative and Convective Heat Transfer from Fire Dynamics Data
- [C.6] Hoseok Jung*, Jiyeon Lee, Sungmo Lee, **Nahyun Lee**, Seungwon Hu, Boyoung Kim, Okkeun Lee, Hyochang Kim, and Hyunmin Kang (2026).
A Wearable Encountered-Type ThermalKinesthetic Interface for Mars Exploration VR
28th International Conference on Human-Computer Interaction (HCII 2026 Invited Papers)
- [P.1] **Nahyun Lee*** and Hyeon Jeon (2026).
Is This Enough to Decide? Facilitating Informed Investment Decisions by Visualizing Missing Evidence
ACM Conference on Human Factors in Computing Systems Extended Abstracts (CHI Extended Abstract 2026)
- [C.5] Myeongjun Kim*, Seungmin Lee*, Junjae Park*, **Nahyun Lee***, and Jo Woon Chong (2025).
A Low-Cost Quantitative Evaluation Testbed of Small-Sized IMU Sensor Modules Using a Smartphone
10th International Conference on Consumer Electronics Asia (ICCE-Asia 2025)

- [C.4] Hoseok Jung*, Jiyeon Lee, Sungmo Lee, and **Nahyun Lee** (2025).
Encounter-type Force and Thermal Haptics for VR with a Wrist-Worn Articulated Robot
Korea Society of Design Science, 2025 Fall International Conference (KSDS 2025)
- [C.3] Hoseok Jung*, Dongho Shin*, **Nahyun Lee***, Minseo Yu*, Kyuhyoung Cho*, and Gyungsun Go* (2024).
Mars Exploration Simulation with an Encounter-type Multimodal Haptic Device
Korea Society of Design Science, 2024 Spring International Conference (KSDS 2024)
- [C.2] Jiyeon Lee*, Jiwon Yang, ChungHyun Lee, **Nahyun Lee**, Hyunmin Kang, and Hoseok Jung (2024).
Proposal for a Virtual Reality Mars Exploration Exhibit Utilizing Reverse-Temperature Multi-Modal Haptic Devices for Popularizing Space Science
International Symposium of Science Museums 2024 (ISSM 2024)
- [C.1] Doohwan Kim*, Yohan Han, Inhyuk Jeong, Yeongseok Hwang, Jinju Park, **Nahyun Lee**, and Yujin Lee (2024).
LLM-based chatbot system to improve worker efficiency and prevent safety incidents
Korea Society of Computer Information, 2024 Winter Conference (KSCI 2024)

ACADEMIC SERVICES

REVIEWER

- ACM International Joint Conference on Pervasive and Ubiquitous Computing (Ubicomp 2026)
- ACM SIGCHI Conference on Human Factors in Computing Systems (CHI 2026)

STUDENT VOLUNTEERS

- IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2025)

EXPERIENCE

INDUSTRY EXPERIENCE

- **Modly** *Feb 2025 - Jan 2026*
Suwon, South Korea
Co-Founder
 - Selected for the Korean I-Corps program (Ministry of Science and ICT); trained at George Washington University
 - Led business strategy for a 3D scanning service, conducting 100+ user interviews to identify viable applications
- **AiGENDA**  *Jun 2024 – Aug 2024*
Suwon, South Korea
AI Developer
 - Developed a lecture-based video Q&A system using STT, LLMs, and LangChain
 - Enhanced domain-specific terminology recognition (60% → 98%) using RAG and vector databases

TEACHING EXPERIENCE

- **Tutoring Program for International Students – Data Structures** *Sep 2025 – Dec 2025*
Suwon, South Korea
Sungkyunkwan University
 - Provided group tutoring for international students enrolled in Data Structures (SWE2015)
- **XR Systems & AI Integration Seminar Lead** *Jul 2025 – Sep 2025*
Suwon, South Korea
Building Information Science & Technology Lab (BIST), Sungkyunkwan University
 - Taught XR development (Unity, Quest, AI pipelines) and guided application in architectural engineering

LEADERSHIP EXPERIENCE

- **Korean I-Corps Project** *Jun 2025 – Jul 2025*
Washington, DC, USA
Ministry of Science and ICT & National Research Foundation
 - Completed training in technology commercialization and venture development at George Washington University
- **President – KingoSmartThings (AI-IoT Entrepreneurship Student Organization)** *Dec 2024 – Feb 2026*
Suwon, South Korea
Sungkyunkwan University
 - Organized campus-wide hackathons (30+ participants) in collaboration with SKKU RISE Center and ICT College
- **Sungkyunkwan University Representative – Young Engineers Honor Society** *Dec 2024 – Dec 2025*
Seoul, South Korea
National Academy of Engineering of Korea
 - Selected as one of three students at Sungkyunkwan University for engineering excellence and leadership

SCHOLARSHIPS

- **Korea-U.S. Advanced Technology Youth Exchange Support Program (~\$18,000)** *Jun 2026 – Present*
Korea Institute for Advancement of Technology
- **Presidential Science Scholarship (~\$18,500)** *Jul 2025 – Present*
Ministry of Science and ICT
- **Cho Byung-Doo Scholarship (~\$5,400)** *Apr 2025 – Sep 2025*
Cho Byung-Doo Scholarship Foundation
- **Korea Entegris Scholarship (~\$2,300)** *Sep 2024*
Entegris
- **Hyeong-Ae Scholarship (~\$4,600)** *Mar 2024 – Sep 2024*
Hyeong-Ae Scholarship Foundation
- **Top 3% Academic Excellence Scholarship (~\$4,200)** *Sep 2023 – Feb 2024*
Sungkyunkwan University
- **Student Success Creativity Scholarship (~\$3,500)** *May 2023*
Sungkyunkwan University

HONORS AND AWARDS

- **Award – 2026 Undergraduate Student Success Story Contest** *June 2026*
Sungkyunkwan University
- **Excellence Award – Excellent Study Club Program** *Feb 2026*
Sungkyunkwan University
- **President's Award – URP III (Undergraduate Interdisciplinary Research Program)** *Dec 2025*
Institute for Cross-disciplinary Studies, Sungkyunkwan University
- **Excellence Award – Embedded Software Competition (Open Innovation Track)** *Nov 2024*
Ministry of Trade, Industry and Energy & Korea Electronics Technology Institute
- **Engineering Innovation Award – Samsung Medical Service Design Program** *Jan 2024*
Bio-Digital Healthcare Consortium, Sungkyunkwan University

ADDITIONAL INFORMATION

Languages: Korean (Native), English (Full professional)

Interests: Birdwatching, Financial Planning, Long-Term Strategy